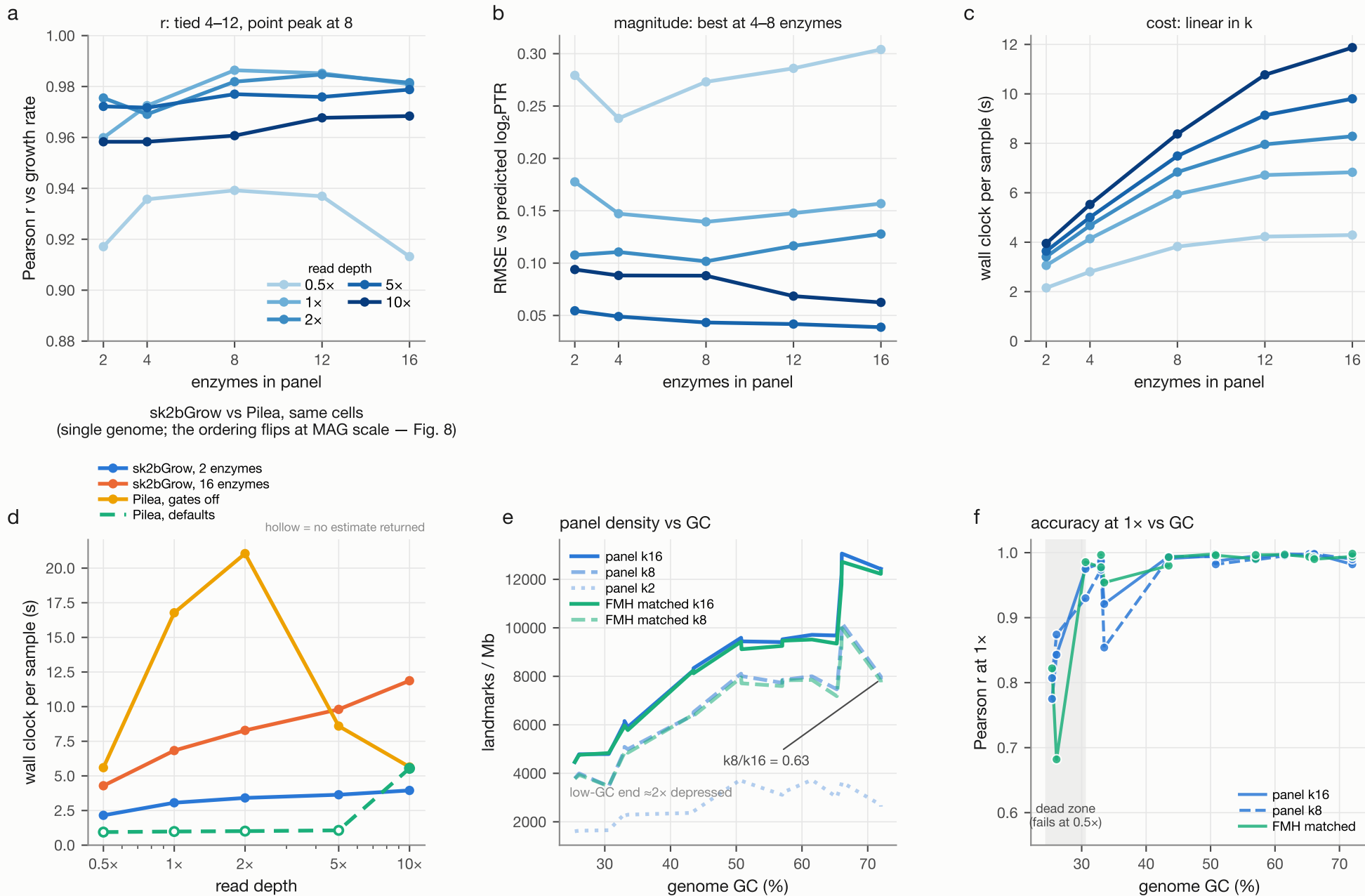




Eighteen real genomes spanning GC 25.4–72.0%, simulated reads, planted  $\log_2\text{PTR}$  0.5–2.0 ( $n = 8$  cells per point). e, Enzyme-panel landmark density against GC: k16 rises monotonically ( $\approx 4.5\text{k}/\text{Mb}$  at 26% to  $\approx 12.4\text{k}/\text{Mb}$  at 72%) and only the low-GC end is depressed ( $\approx 2\times$  below mid-GC); density-matched FracMinHash tracks the panel within a few percent at every GC. The k8/k16 ratio thins at the GC extremes (0.63 at 72%) — the extra eight enzymes buy high-GC density, not accuracy. f, Pearson  $r$  at 1x: k8 matches k16 at every GC despite carrying only 63% of its landmarks at 72% GC (A6 holds across GC), and matched FracMinHash ties the panel. Shaded band: the instrument boundary — at GC  $\leq 30\%$  together with 0.5x depth every landmark mode fails ( $r < 0.7$ ) — so the claimable range is GC  $\geq 30\%$  at  $\geq 1\times$  coverage.